Handling of material junctions in SCUFF-EM

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1 Material Junctions

This is distinct

2 Half-RWG Basis Functions

The half-RWG (HRWG) basis function associated with an edge $\mathbf{L}_\alpha$ of a panel $\mathcal{P}$ is

$$\mathbf{h}_\alpha(\mathbf{x}) = \begin{cases} \dfrac{l_\alpha}{2A_\alpha}(\mathbf{x}-\mathbf{Q}_\alpha), & \mathbf{x}\in\mathcal{P} \\ 0, & \text{otherwise} \end{cases}$$

where l_α is the edge length, $\mathbf{Q}_\alpha$ is the vertex of $\mathcal{P}$ opposite the edge, and A_α is the area of $\mathcal{P}$. Introducing the *indicator function* $\chi_\mathcal{P}(\mathbf{x})$, which equals unity for $\mathbf{x}\in\mathcal{P}$ and 0 otherwise, we can rewrite this in the form

$$\mathbf{h}_\alpha(\mathbf{x}) = \frac{l_\alpha}{2A_\alpha}(\mathbf{x}-\mathbf{Q}_\alpha)\chi_\mathcal{P}(\mathbf{x}).$$

The novelty of the HRWG basis function is that the charge density associated with it has both a surface-charge portion (like the ordinary RWG basis function) and a new *line-charge* density on edge $\mathbf{L}_\alpha$. Suppose we have an HRWG function $\mathbf{h}_\alpha$ populated with strength k_α. (Recall k_α has units of current/length.) Then the surface and line charge densities associated with this current distribution are

$$\sigma(\mathbf{x}) = \frac{lk_\alpha}{i\omega A}\chi_\mathcal{P}(\mathbf{x})$$
$$\lambda(\mathbf{x}) = -\frac{k_\alpha}{i\omega}\chi_{\mathbf{L}_\alpha}(\mathbf{x})$$

3 Matrix Elements involving HRWG Basis Functions

3.1 Matrix elements between RWG and HRWG functions

Consider an RWG function $\mathbf{f}_\alpha$ and an HRWG function $\mathbf{h}_\beta$. $\mathbf{f}_\alpha$ describes surface currents on two panels $\mathcal{P}_\alpha^\pm$, while $\mathbf{h}_\beta$ describes surface currents on a single panel $\mathcal{P}_\beta$ plus the line charge on an edge $\mathbf{L}_\beta$. The BEM matrix elements between these two basis functions involve the interactions

$$\left\langle \mathbf{f}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle = \left\langle \mathbf{f}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–panel}} + \left\langle \mathbf{f}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–edge}} \tag{1a}$$

$$\left\langle \mathbf{f}_\alpha \middle| \mathbf{C} \middle| \mathbf{h}_\beta \right\rangle = \left\langle \mathbf{f}_\alpha \middle| \mathbf{C} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–panel}} \tag{1b}$$

The first terms here are just the usual panel-panel interactions between currents on panels $\mathcal{P}_\alpha^\pm$ and $\mathcal{P}_\beta$, and may be computed using the usual LIBSCUFF methods for panel-panel interactions.

The second term in (1a) is a new interaction between the surface charge densities on $\mathcal{P}_\alpha^\pm$ and the line charge density on $\mathbf{L}_\beta$:

$$\left\langle \mathbf{f}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–edge}} = -\frac{1}{k^2} \sum \pm \int_{\mathcal{P}_\alpha^\pm} d\mathbf{x} \int_{\mathbf{L}_\beta} d\mathbf{x}' \left\{ \Big[\nabla \cdot \mathbf{f}_\alpha(\mathbf{x}) \Big] G(\mathbf{x},\mathbf{x}') \Big[\nabla \cdot \mathbf{h}_\alpha(\mathbf{x}') \Big] \right\} \tag{2}$$

Note that there is no panel-edge contribution to $\langle \mathbf{f} | \mathbf{C} | \mathbf{h} \rangle$.

3.2 Matrix elements between HRWG and HRWG functions

Next consider an HRWG function $\mathbf{h}_\alpha$ and an HRWG function $\mathbf{h}_\beta$. We have

$$\left\langle \mathbf{h}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle = \left\langle \mathbf{h}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–panel}} + \left\langle \mathbf{h}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–edge}} \tag{3a}$$

$$\left\langle \mathbf{h}_\alpha \middle| \mathbf{C} \middle| \mathbf{h}_\beta \right\rangle = \left\langle \mathbf{h}_\alpha \middle| \mathbf{C} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–panel}} \tag{3b}$$

As above, the panel-panel interactions may be computed using the usual LIBSCUFF methods for panel-panel interactions.

The second term in (3a) is similar to (**??**), but without the sum over positive and negative panels, since there is only one panel in the support of $\mathbf{h}_\alpha$:

$$\left\langle \mathbf{h}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–edge}} = -\frac{1}{k^2} \int_{\mathcal{P}_\alpha} d\mathbf{x} \int_{\mathbf{L}_\beta} d\mathbf{x}' \left\{ \Big[\nabla \cdot \mathbf{h}_\alpha(\mathbf{x}) \Big] G(\mathbf{x},\mathbf{x}') \Big[\nabla \cdot \mathbf{h}_\alpha(\mathbf{x}') \Big] \right\}$$

Nonsingular Edge-Panel Interactions

$$\left\langle \mathbf{f}_\alpha \middle| \mathbf{G} \middle| \mathbf{h}_\beta \right\rangle^{\text{panel–edge}} = -\frac{1}{k^2} \sum \pm \frac{l_\alpha}{A_\alpha^\pm} \int_{\mathcal{P}_\alpha^\pm} d\mathbf{x} \int_{\mathbf{L}_\beta} d\mathbf{x}' G(\mathbf{x},\mathbf{x}')$$

Consider one of the two integrals here [Figure **??(a)**]:

$$\begin{aligned} &-\frac{l_\alpha}{Ak^2} \int_{\mathcal{P}} d\mathbf{x} \int_{\mathbf{L}_\beta} d\mathbf{x}' G(\mathbf{x},\mathbf{x}') \\ &\quad = -\frac{2 l_\alpha l_\beta}{k^2} \int_0^1 du \int_0^u dv \int_0^1 du' G\Big[\mathbf{x}\big(u,v\big), \mathbf{x}'\big(u'\big)\Big] \qquad (4) \\ &\qquad \mathbf{x} = \mathbf{V}_0 + u\mathbf{A} + v\mathbf{B}, \qquad \mathbf{x}' = \mathbf{V}_0' + u'\mathbf{L} \end{aligned}$$

This integral is easy to evaluate via three-dimensional numerical cubature.

Singular Edge-Panel Interactions: The Common Vertex Case

If the edge or panel share a single vertex [Figure **??(bc)**], then the integral is singular and we have to be more clever. In this case we parameterize points in $\mathcal{P}$ and $\mathbf{L}$ according to

$$\mathbf{x} = \mathbf{V}_0 + u\mathbf{A} + v\mathbf{B} \qquad \mathbf{x}' = \mathbf{V}_0 + u'\mathbf{L},$$

and it is shown in Appendix A that the singular three-dimensional integral in (4) may be reduced to a nonsingular two-dimensional integral:

$$\begin{aligned} &-\frac{l_\alpha}{Ak^2} \int_{\mathcal{P}} d\mathbf{x} \int_{\mathbf{L}_\beta} d\mathbf{x}' G(\mathbf{x},\mathbf{x}') \qquad (5) \\ &\quad = -\frac{l_\alpha l_\beta}{4\pi k^2} \int_0^1 dx \int_0^1 dy \Big\{ \quad \frac{x}{R_1} e^{ikR_1} \cdot \text{ExpRel}(2, -ikR_1) \qquad (6) \\ &\qquad\qquad + \frac{(1-x)}{R_2} e^{ikR_2} \cdot \text{ExpRel}(2, -ikR_2) \qquad (7) \\ &\qquad\qquad + \frac{x}{R_3} e^{ikR_3} \cdot \text{ExpRel}(2, -ikR_3) \Big\} \qquad (8) \end{aligned}$$

with

$$R_1(x,y) = \Big|x\mathbf{A} + xy\mathbf{B} - \mathbf{L}\Big|, \qquad R_2(x,y) = \Big|\mathbf{A} + (1-x)y\mathbf{B} - x\mathbf{L}\Big| \qquad R_3(x,y) = \Big|\mathbf{A} + (1-xy)\mathbf{B} - x\mathbf{L}\Big|$$

and

$$\text{ExpRel}(N, X) \equiv \frac{N!}{X^N}\Big[e^X - 1 - X - \frac{X^2}{2!} - \cdots - \frac{X^{N-1}}{(N-1)!}\Big].$$

A Taylor-Duffy transformation of edge-panel integrals

A.1 The common vertex case

In this section we justify equation (**??**).

The integration region for the integral on the LHS of (**??**) is the direct product of a unit right triangle in the (u, v) plane with the unit interval on the u' axis, i.e.

$$\text{region of integration} = \mathcal{R} = \begin{cases} 0 \le u \le 1 \\ 0 \le v \le u \\ 0 \le u' \le 1. \end{cases}$$

Step 1. Decompose region of integration into tetrahedra

Following the spirit of the Taylor-Duffy method, we decompose $\mathcal{R}$ as the non-intersecting union of three tetrahedral regions:

$$\mathcal{R} = \mathcal{R}_1 \cup \mathcal{R}_2 \cup \mathcal{R}_3$$

where

$$\mathcal{R}_1 = \begin{cases} 0 \le u' \le 1 \\ 0 \le u \le u' \\ 0 \le v \le u \end{cases}, \qquad \mathcal{R}_2 = \begin{cases} 0 \le u \le 1 \\ 0 \le u' \le u \\ 0 \le v \le u - u' \end{cases} \qquad \mathcal{R}_3 = \begin{cases} 0 \le u \le 1 \\ 0 \le u' \le u \\ u - u' \le v \le u. \end{cases}$$

Step 2. Introduce Duffy-transformed variables

The next step is to introduce Duffy-transformed variables for each of the three integration regions. The idea of the Duffy transform is to single out one of the three variables as the "independent" variable (call it w), and then to "measure the other two variables in units of w." This ensures that the transformed versions of all three of the original variables u, v, u' are now expressed as multiples of w, whereupon a factor of w may be pulled out of radicals like $\sqrt{Au^2 + Bv^2 + Cu'^2}$. When such radicals appear in the denominator of the integrand, we then get overall multiplicative factors of w in the denominator, which cancel with the Jacobian of the variable transformation to yield

It is also convenient, although not strictly necessary, to define our Duffy-transformed variables w, x, y in such a way that the region of integration for all three of the transformed integrals is the unit three-dimensional cube (i.e. w, x, y all running from 0 to 1.) This allows the three integrals to be combined into a *single* integral over the unit cube.

Such a variable transformation is

$$\begin{aligned}
\mathcal{R}_1: &\quad u'=w, &\quad u=wx, &\quad v=wxy, &\quad \mathcal{J}=w^2x\\
\mathcal{R}_2: &\quad u=w, &\quad u'=wx, &\quad v=w(1-x)y, &\quad \mathcal{J}=w^2(1-x)\\
\mathcal{R}_3: &\quad u=w, &\quad u'=wx, &\quad v=w-wxy, &\quad \mathcal{J}=w^2x
\end{aligned}$$

and our integral in terms of the transformed variables becomes

$$\begin{aligned}
&\int_0^1 du \int_0^u dv \int_0^1 du' f(u,v,u') \\
&\quad = \int_0^1 dw \int_0^1 dx \int_0^1 dy \Big\{ \quad w^2 x f\Big(wx, wxy, w\Big) \\
&\qquad\qquad\qquad\qquad + w^2(1-x) f\Big(w, w(1-x)y, wx\Big) \\
&\qquad\qquad\qquad\qquad + w^2 x f\Big(w, w(1-xy), wx\Big) \Big\}
\end{aligned} \tag{9}$$

Step 3. Evaluate the w integral

The integrand in question here is

$$f(u,v,u') = \frac{e^{ikR(u,v,u')}}{4\pi R(u,v,u')}, \qquad R(u,v,u') = \Big|u\mathbf{A} + v\mathbf{B} - \mathbf{u}'\mathbf{L}\Big|$$

(where $\mathbf{A}=\mathbf{L}$ for the shared-edge case). Applying (9), we have

$$\begin{aligned}
&\int_0^1 du \int_0^u dv \int_0^1 du' f(u,v,u') \\
&= \frac{1}{4\pi} \int_0^1 dx \int_0^1 dy \int_0^1 dw \Big[\frac{wxe^{ikwR_1(x,y)}}{R_1(x,y)} + \frac{w(1-x)e^{ikwR_2(x,y)}}{R_2(x,y)} + \frac{wxe^{ikwR_3(x,y)}}{R_3(x,y)} \Big]
\end{aligned} \tag{10}$$

$$R_1(x,y) = \Big|x\mathbf{A}+xy\mathbf{B}-\mathbf{L}\Big|, \qquad R_2(x,y) = \Big|\mathbf{A}+(1-x)y\mathbf{B}-x\mathbf{L}\Big| \qquad R_3(x,y) = \Big|\mathbf{A}+(1-xy)\mathbf{B}-x\mathbf{L}\Big|.$$

The w integral may be evaluated in closed form:

$$\int_0^1 we^{wX} dw = \frac{1}{2} e^X \cdot \mathrm{ExpRel}(2,-X) \tag{11}$$

where $\mathrm{ExpRel}(N,X)$ is e^X, minus the first N terms in its series expansion, normalized so that ExpRel(N,0) = 1:

$$\begin{aligned}
\mathrm{ExpRel}(N,X) &\equiv \frac{N!}{X^N}\Big[e^X - 1 - X - \frac{X^2}{2!} - \cdots - \frac{X^{N-1}}{(N-1)!}\Big] \\
&= \frac{N!}{X^N}\Big[\frac{X^N}{N!} + \frac{X^{N+1}}{(N+1)^1} + \cdots\Big] \\
&= 1 + \frac{X}{N+1} + \frac{X^2}{(N+1)(N+2)} + \cdots
\end{aligned}$$

Using (11), equation (10) becomes

$$= \frac{1}{8\pi}\int_0^1 dx \int_0^1 dy \Bigg\{ \frac{x}{R_1} e^{ikR_1} \cdot \mathrm{ExpRel}(2, -ikR_1) + \frac{(1-x)}{R_2} e^{ikR_2} \cdot \mathrm{ExpRel}(2, -ikR_2) + \frac{x}{R_3} e^{ikR_3} \cdot \mathrm{ExpRel}(2, -ikR_3) \Bigg\}$$

where R_1, R_2, R_3 are functions of x and y. This is a nonsingular integral over the unit square which may be evaluated by straighforward numerical cubature.

A.2 The common edge case

In this case we parameterize points in the triangle and on the edge according to

$$\mathbf{x} = \mathbf{V}_0 + u\mathbf{A} + v\mathbf{B}, \qquad \mathbf{x}' = \mathbf{V}_0 + u\mathbf{A}.$$

The edge-panel integral is

$$\texttt{EPI} \equiv \int_0^1 du' \int_0^1 du \int_0^u dv\, \mathcal{I}(u-u', v) \tag{12}$$

$$\mathcal{I}(u-u', v) = \frac{e^{ikR(u-u',v)}}{4\pi R(u-u', v)}, \qquad R(u-u', v) = \Big|(u-u')\mathbf{A} + v\mathbf{B}\Big|$$

We begin by rewriting the integral over the triangle in the u, v plane: for any integrand function $F(u, v)$, we have

$$\int_0^1 du \int_0^u dv\, F(u, v) = \int_0^1 dv \int_v^1 du\, F(u, v)$$

Applying this to (12) yields

$$\texttt{EPI} = \int_0^1 dv \int_0^1 du' \int_y^1 du\, \mathcal{I}(u-u', v). \tag{13}$$

Now we rewrite the u, u' integral to take advantage of the fact that the integrand only depends on the difference $u-u'$. Putting $s = (u+u'), t = u-u'$, the result we need (again valid for arbitrary integrand functions $g(u, u')$ is

$$\int_0^1 du' \int_v^1 du\, g(u, u')$$
$$= \frac{1}{2}\Bigg\{ \int_{v-1}^0 dt \int_{2v-t}^2 ds \;+\; \int_0^v dt \int_{2v-t}^{2-t} ds \;+\; \int_v^1 dt \int_t^{2-t} ds \Bigg\} g\Big(\frac{1}{2}(s+t), \frac{1}{2}(s-t)\Big).$$

Applying this to (13) and evaluating the s integrals allows us to express the EPI as a sum of three two-dimensional integrals:

$$\texttt{EPI} = \texttt{EPI}_1 + \texttt{EPI}_2 + \texttt{EPI}_3$$

$$\texttt{EPI}_1 = \frac{1}{2}\int_0^1 dv \int_{v-1}^0 dt\ (2+t-2v)\mathcal{I}(t,v)$$

$$\texttt{EPI}_2 = \frac{1}{2}\int_0^1 dv \int_0^v dt\ (2-2v)\,\mathcal{I}(t,v)$$

$$\texttt{EPI}_3 = \frac{1}{2}\int_0^1 dv \int_v^1 dt\ (2-2t)\,\mathcal{I}(t,v).$$

Now make Duffy transformations:

$$\begin{aligned}
\texttt{EPI}_1 \quad &\xrightarrow[v=xy]{t=x(y-1)} \quad \frac{1}{2}\int_0^1 dx \int_0^1 dy\ x(2-xy-x)\,\mathcal{I}\Big(x(y-1),xy\Big)\\
\texttt{EPI}_2 \quad &\xrightarrow[y=t/v]{x=v} \quad \frac{1}{2}\int_0^1 dx \int_0^1 dy\ x(2-2x)\,\mathcal{I}\Big(xy,x\Big)\\
\texttt{EPI}_3 \quad &\xrightarrow[y=v/t]{x=t} \quad \frac{1}{2}\int_0^1 dx \int_0^1 dy\ x(2-2x)\,\mathcal{I}\Big(x,xy\Big)
\end{aligned}$$

Recombine into a single big integral:

$$\texttt{EPI} = \frac{1}{8\pi}\int_0^1 dx \int_0^1 dy \left\{(2-xy-x)\frac{e^{ikxR_1(y)}}{R_1(y)} + (2-2x)\frac{e^{ikxR_2(y)}}{R_2(y)} + (2-2x)\frac{e^{ikxR_3(y)}}{R_3(y)}\right\}$$

$$R_1(y) = \Big|(y-1)\mathbf{A} + y\mathbf{B}\Big|, \qquad R_2(y) = \Big|y\mathbf{A} + \mathbf{B}\Big|, \qquad R_3(y) = \Big|\mathbf{A} + y\mathbf{B}\Big|.$$

Finally, evaluate the x integral, leaving behind a single one-dimensional integral to be evaluated numerically:

$$\begin{aligned}
\texttt{EPI} = \frac{1}{8\pi}\int_0^1 dy \Bigg\{ & \frac{e^{ikR_1}}{R_1}\Big[2\,\texttt{ExpRel}\big(1,-ikR_1\big) - \frac{y+1}{2}\texttt{ExpRel}\big(2,-ikR_1\big)\Big]\\
&+\frac{e^{ikR_2}}{R_2}\Big[2\,\texttt{ExpRel}\big(1,-ikR_2\big) - \texttt{ExpRel}\big(2,-ikR_2\big)\Big]\\
&+\frac{e^{ikR_3}}{R_3}\Big[2\,\texttt{ExpRel}\big(1,-ikR_3\big) - \texttt{ExpRel}\big(2,-ikR_3\big)\Big]\Bigg\}.
\end{aligned}$$